

DEVI : AI Based Domestic Violence Detection and Risk Assessment System

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Abstract— Domestic violence is a critical social issue that often remains underreported, making timely intervention difficult. This paper presents DEVI (Domestic Violence Detection and Risk Assessment using Artificial Intelligence), an AI-based system designed to analyze domestic violence data and predict potential risk using machine learning techniques. The proposed framework includes data preprocessing, exploratory data analysis, feature engineering, and model training on crime-related datasets. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, are evaluated using standard performance metrics such as Accuracy, Precision, Recall, and F1-Score. The system also provides visual analytics to identify crime patterns and trends, supporting informed decision-making for law enforcement agencies, researchers, and policymakers. The proposed approach demonstrates the potential of artificial intelligence to enhance domestic violence analysis and enable data-driven risk assessment for improved prevention and response.

Keywords— Domestic Violence, Artificial Intelligence, Machine Learning, Crime Analytics, Risk Assessment, Predictive Analytics.

I. INTRODUCTION

A. Background

Domestic violence is one of the most pervasive forms of violence worldwide, affecting individuals regardless of age, socioeconomic status, or cultural background. It encompasses physical, emotional, psychological, sexual, and economic abuse, leading to long-term consequences for victims and society. According to reports from the World Health Organization (WHO) and the National Crime Records Bureau (NCRB), a significant number of domestic violence cases remain unreported due to fear, social stigma, financial dependence, and limited access to support systems. This underreporting poses a major challenge to timely intervention and victim protection.

With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML), data-driven techniques have emerged as powerful tools for analysing complex patterns within large datasets. These technologies enable the identification of hidden trends, prediction of potential risks, and generation of actionable insights that are difficult to obtain through traditional analytical methods. In the context of domestic violence, AI can assist in understanding crime

patterns, identifying high-risk scenarios, and supporting authorities in making informed decisions.

The **DEVI (Detection and Evaluation of Violence Indicators)** framework is proposed to address this challenge by utilizing machine learning techniques for domestic violence analysis and prediction. The system integrates crime-related datasets, performs comprehensive data preprocessing and exploratory analysis, and evaluates multiple machine learning algorithms to identify the most effective predictive model. Additionally, DEVI provides interactive visualizations, enabling users to analyse crime trends and obtain prediction results in an accessible and user-friendly manner. By leveraging AI-driven analytics, the proposed system aims to contribute toward evidence-based decision-making and enhance efforts in domestic violence prevention and risk assessment.

B. Problem Statement

Domestic violence remains one of the most underreported forms of violence despite its widespread social and psychological impact. Victims often hesitate to report incidents due to fear, social stigma, financial dependence, or lack of timely support, making early identification and intervention extremely difficult. As the volume of crime-related data continues to increase, traditional methods of data analysis become inefficient in identifying hidden patterns and emerging trends.

Existing approaches primarily rely on manual investigation and retrospective analysis, which are time-consuming and often fail to provide predictive insights. While recent studies have explored the application of machine learning in crime analysis, many focus on a limited set of algorithms or specific datasets, resulting in reduced generalizability and limited comparative evaluation. Furthermore, most existing systems do not integrate comprehensive data preprocessing, exploratory data analysis, predictive modelling, and visualization within a single framework.

Therefore, there is a need for an intelligent, data-driven system capable of analysing domestic violence data, identifying meaningful patterns, and predicting potential risk more effectively. Such a system can support researchers, policymakers, and law enforcement agencies by providing

accurate analytical insights, improving decision-making, and contributing to more proactive prevention strategies.

C. Objective

The primary objective of this research is to develop an intelligent framework for analysing domestic violence-related data using machine learning techniques to support accurate prediction and data-driven decision-making. The study aims to improve the understanding of crime patterns and evaluate the effectiveness of different predictive models.

The specific objectives of this research are as follows:

- To collect and preprocess domestic violence and crime-related datasets for effective analysis.
- To perform exploratory data analysis (EDA) to identify significant trends, patterns, and relationships within the data.
- To develop and compare multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, for domestic violence prediction.
- To evaluate the performance of the developed models using standard metrics such as Accuracy, Precision, Recall, and F1-Score.
- To identify the most suitable predictive model for domestic violence risk assessment based on comparative analysis.

D. Contribution

The major contributions of this research are summarized as follows:

- An intelligent framework is proposed for the analysis and prediction of domestic violence using machine learning techniques.
- A comprehensive workflow is developed that integrates data collection, preprocessing, exploratory data analysis, feature engineering, model training, and performance evaluation.
- Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, are implemented and comparatively evaluated to identify the most effective predictive model.
- Visual analytics are incorporated to present crime trends and patterns through interactive graphs and dashboards, enabling better interpretation of the data.
- The proposed framework aims to support researchers, policymakers, and law enforcement agencies by facilitating data-driven decision-making and promoting the application of artificial intelligence in domestic violence analysis.

II. WORK

The application of Artificial Intelligence (AI) and Machine Learning (ML) in crime analysis has gained significant attention

in recent years due to its ability to process large volumes of data and identify hidden patterns. Researchers have developed various predictive models to support crime analysis and decision-making by utilizing historical crime records and statistical information. These intelligent techniques assist in understanding crime trends, identifying high-risk regions, and improving resource allocation for law enforcement agencies.

Several machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, have been widely employed for crime prediction tasks. Comparative studies have shown that ensemble learning methods, particularly Random Forest and XGBoost, often achieve higher prediction accuracy because of their ability to handle complex and non-linear relationships within datasets. The performance of these models largely depends on the quality of data preprocessing, feature engineering, and model optimization.

Recent studies have also highlighted the importance of data preprocessing and exploratory data analysis in improving predictive performance. Techniques such as handling missing values, removing duplicate records, feature encoding, normalization, and visualization help in enhancing data quality and identifying significant relationships among different variables. Graphical analyses, including trend analysis, correlation matrices, and distribution plots, provide meaningful insights into crime patterns and support informed decision-making.

In the domain of domestic violence, several research efforts have focused on analysing textual complaints, social media posts, and case reports using Natural Language Processing (NLP) and deep learning techniques. While these approaches demonstrate promising results for text classification, they generally concentrate on a single data source and provide limited support for comprehensive crime analytics or comparative evaluation of multiple machine learning models.

Although previous studies have demonstrated the effectiveness of artificial intelligence in crime prediction, many existing systems are limited in terms of dataset diversity, model comparison, and practical deployment. Most approaches focus either on prediction or visualization rather than integrating data preprocessing, exploratory analysis, predictive modelling, performance evaluation, and user-friendly deployment into a single framework. These limitations motivate the development of a more comprehensive system capable of providing accurate predictions along with interactive visual analytics for domestic violence data.

TABLE 1 (Papers Studied)

Ref.	Source	Methodology	Key Findings	Limitation
[1]	Using Machine Learning Prediction to Create a 15-question IPV	Machine Learning (NFHS dataset)	Developed a 15-question tool to identify women at high risk of intimate	Focuses only on screening and not comprehensive crime analytics.

Ref.	Source	Methodology	Key Findings	Limitation
	<i>Measurement Tool</i> (Journal of Interpersonal Violence, 2024)		partner violence with strong recall.	
[2]	<i>Predicting Domestic Abuse (Fairly) and Police Risk Assessment</i> (2023)	Logistic Regression with Police Data	Improved domestic abuse risk prediction over traditional DASH assessment using police records.	Uses UK police data, limiting applicability to other regions.
[3]	<i>Utilizing Text Mining, Data Linkage and Deep Learning in Police and Health Records to Predict Future Offenses in Family and Domestic Violence</i> (Frontiers in Digital Health, 2021)	Deep Learning + Text Mining	Combined police and health records to predict future domestic violence offenses.	Requires linked institutional datasets that may not always be available.
[4]	<i>Using Machine Learning to Forecast Domestic Homicide via Police Data and Super Learning</i> (Scientific Reports, 2023)	Super Learner Ensemble	Improved domestic homicide forecasting using multiple machine learning models.	Focused specifically on homicide prediction rather than broader domestic violence analysis.
[5]	<i>Using Unsupervised Machine Learning to</i>	Unsupervised Learning (Clustering)	Identified distinct perpetrator profiles from	Performs clustering but not predictive classification.

Ref.	Source	Methodology	Key Findings	Limitation
	<i>Find Profiles of Domestic Abuse Perpetrators</i> (Policing: A Journal of Policy and Practice, 2024)		longitudinal police datasets.	
[6]	<i>Comparing Conventional and Machine-Learning Approaches to Risk Assessment in Domestic Abuse Cases</i> (Journal of Empirical Legal Studies, 2021)	Machine Learning Risk Assessment	Demonstrated that ML-based risk assessment outperformed conventional protocols.	Primarily focused on police risk assessment.
[7]	<i>Crime Prediction Using Machine Learning and Deep Learning: A Systematic Review and Future Directions</i> (2023)	Systematic Review	Reviewed more than 150 studies and summarized datasets, algorithms, and future research directions.	Review paper; no implementation or experimental validation.
[8]	<i>Deep Learning Based Crime Prediction Models: Experiments and Analysis</i> (2024)	Deep Learning Comparison	Compared major deep learning models for crime prediction and recommended suitable architectures.	Focused on general crime prediction rather than domestic violence.
[9]	<i>Analysing Domestic Violence through Exploratory</i>	Ensemble Learning + Explainable AI	Combined EDA, ensemble learning, SHAP, and	Study focuses on a specific demographic and dataset.

Ref.	Source	Methodology	Key Findings	Limitation
	<i>Data Analysis and Explainable Ensemble Learning Insights (2024)</i>		LIME to improve domestic violence prediction and interpretability.	
[10]	National Crime Records Bureau (NCRB), Crime in India Reports	Statistical Crime Analysis	Provides comprehensive official crime statistics used for policy formulation and research.	Does not include predictive analytics or AI-based analysis.

A. Research Gap

The literature review indicates that existing studies have successfully applied machine learning and deep learning techniques for domestic violence detection, risk assessment, and crime prediction. However, most research focuses on specific objectives such as intimate partner violence screening, police risk assessment, text-based classification, or domestic homicide prediction. These studies are generally limited to a single dataset, a specific algorithm, or a particular application domain, reducing their adaptability to broader domestic violence analysis.

Furthermore, several studies emphasize predictive modelling without providing a comprehensive framework that integrates data preprocessing, exploratory data analysis (EDA), comparative evaluation of multiple machine learning algorithms, and interactive visualization. Although explainable AI and ensemble learning have improved model interpretability, user-friendly deployment for real-time analysis and decision support remains limited. Official crime reports, such as those published by the National Crime Records Bureau (NCRB), provide valuable statistical information but do not incorporate predictive analytics or intelligent decision-support mechanisms. To address these limitations, the proposed research develops a comprehensive framework for domestic violence analysis by integrating data preprocessing, exploratory data analysis, feature engineering, and comparative evaluation of multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost. This approach aims to support researchers, policymakers, and law enforcement agencies with accurate, data-driven insights for domestic violence analysis and risk assessment.

TABLE 2 (Research Gap Analysis)

Existing Studies	Identified Gap	Proposed Work
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Focus on a single algorithm or prediction technique	Limited comparative evaluation of machine learning models	Comparative analysis of Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost
Use of specific datasets such as police records, text complaints, or health data	Limited integration of diverse crime-related data	Comprehensive analysis of domestic violence and crime datasets
Emphasis on prediction only	Lack of an end-to-end framework combining preprocessing, EDA, modelling, and evaluation	Integrated pipeline covering preprocessing, EDA, feature engineering, model training, and evaluation
Official crime reports provide descriptive statistics only	No predictive analytics for proactive decision-making	AI-based predictive framework for domestic violence risk assessment

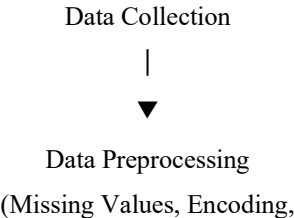
III. PROPOSED METHODOLOGY

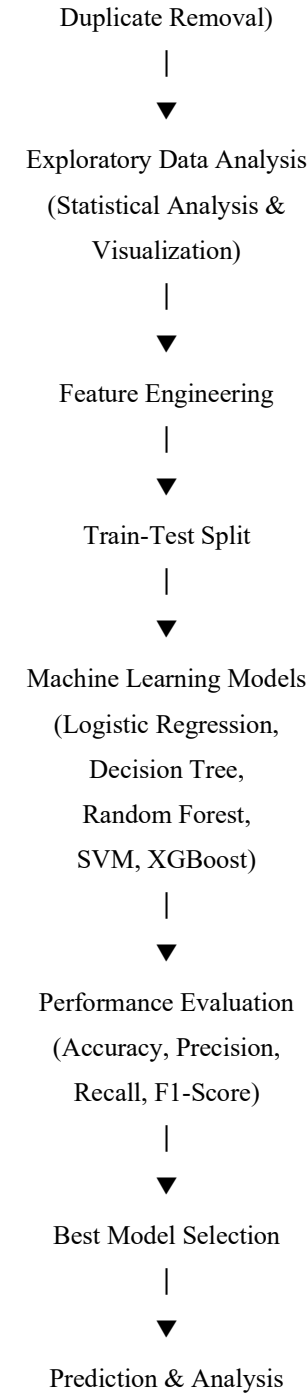
A. System Overview

The proposed methodology presents a systematic framework for analysing and predicting domestic violence using machine learning techniques. The process begins with the collection of crime-related datasets from reliable sources, followed by data preprocessing to improve data quality by handling missing values, removing duplicate records, and encoding categorical features. Exploratory Data Analysis (EDA) is then performed to identify significant trends, patterns, and relationships within the dataset through statistical analysis and visual representations.

After preprocessing, feature engineering techniques are applied to select and transform relevant features that improve the performance of predictive models. The processed dataset is divided into training and testing sets, and multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, are trained and evaluated. The performance of each model is assessed using evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. Based on the comparative analysis, the best-performing model is selected for domestic violence prediction. The proposed framework aims to provide accurate predictions and meaningful insights that can assist researchers and policymakers in understanding crime patterns and supporting data-driven decision-making.

B. System Workflow





IV. DATA DESCRIPTION.

A. Data Sources

The dataset used in this research was obtained from publicly available crime statistics related to crimes against women in India. It contains historical records of various categories of crimes reported across different states and union territories over multiple years. The dataset serves as the foundation for analyzing crime trends and developing predictive machine learning models.

B. Data Overview

The dataset consists of records representing different crime categories reported against women from 2001 to 2022. Each record includes geographical information, the year of occurrence, the type of crime, and the corresponding number of reported cases. The collected data provides sufficient diversity for performing trend analysis, exploratory data analysis, and predictive modelling.

The dataset was examined to understand its structure, identify missing values, remove duplicate entries, and ensure consistency before applying machine learning algorithms.

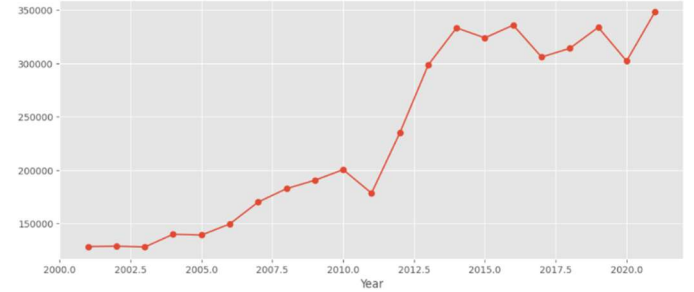


TABLE 3 (Description of Dataset Attributes)

Attribute	Description
State/UT	Name of the State or Union Territory where the crime was reported
Year	Year in which the crime was recorded
Crime Type	Category of crime against women (e.g., Domestic Violence, Kidnapping, Assault, etc.)
Cases Reported	Total number of reported cases for the corresponding crime category

C. Data Characteristics

The dataset contains both categorical and numerical attributes. The State/UT and Crime Type columns are categorical variables, whereas Year and Cases Reported are numerical variables. Such a combination makes the dataset suitable for statistical analysis, visualization, and supervised machine learning after appropriate preprocessing.

D. Data Preprocessing and Exploratory Data Analysis

Before Data preprocessing is an essential step in machine learning, as the quality of input data directly influences the performance of predictive models. Before training the models, the collected dataset was carefully examined to identify inconsistencies, improve data quality, and transform the data into a suitable format for analysis. The following preprocessing techniques were applied.

Handling Missing Values

The dataset was inspected for missing or null values to ensure data completeness. Missing entries, if present, were handled using appropriate techniques such as removal or replacement based on the nature of the attribute. This process helped maintain data consistency and reduced the possibility of inaccurate model predictions caused by incomplete information.

Duplicate Record Removal

Duplicate records can introduce bias and negatively affect model performance. Therefore, the dataset was examined for duplicate entries, and redundant records were removed. This ensured that each observation represented unique information, improving the reliability of the analysis.

Data Cleaning

Data cleaning was performed to eliminate inconsistencies such as incorrect formatting, unnecessary spaces, and invalid entries. The dataset was standardized to maintain uniformity across all features, making it suitable for further processing and machine learning applications.

Feature Encoding

Machine learning algorithms require numerical input; therefore, categorical variables such as state names and crime categories were transformed into numerical representations using appropriate encoding techniques. This conversion enabled the algorithms to effectively process categorical information during model training.

Feature Scaling

Numerical features often have different value ranges, which may affect the performance of certain machine learning algorithms. Feature scaling was applied where necessary to normalize the data and ensure that all numerical attributes contributed equally during model training.

Train-Test Split

After preprocessing, the dataset was divided into training and testing subsets. The training set was used to develop the machine learning models, while the testing set was used to evaluate their predictive performance on previously unseen data. This approach helped assess the generalization capability of the models.

TABLE 4 (Dataset Statistics)

Description	Value
Dataset Name	Crime Against Women (2001–2022)
Time Period	2001–2022
Country	India
Data Type	Structured Tabular Data
Features	State/UT, Year, Crime Category, Cases Reported
Target Variable	Cases Reported

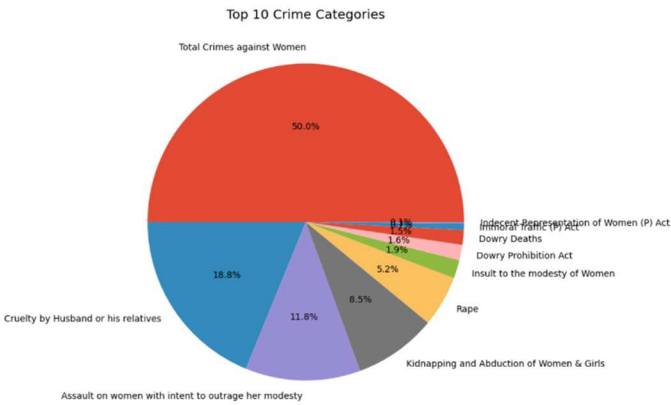
Before Data preprocessing is an essential step in machine learning, as the quality

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the dataset and identify meaningful patterns before developing predictive models. Various statistical and visualization techniques were employed

to analyse the distribution of crime records, detect trends, and examine relationships among different variables.

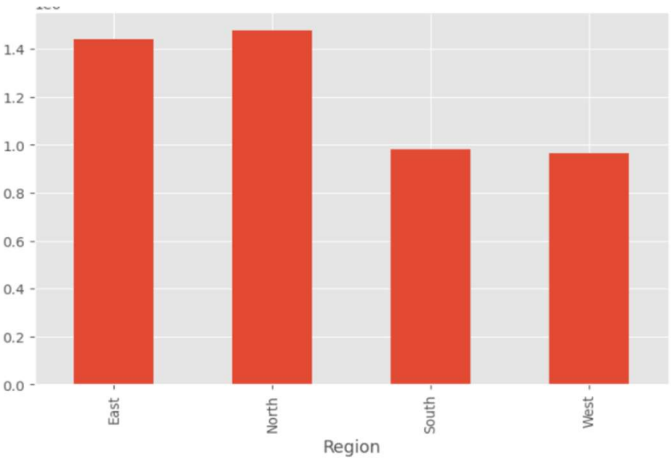
Dataset Overview

Initially, the overall structure of the dataset was examined by analysing the number of records, attributes, and data types. Basic statistical measures such as mean, median, minimum, maximum, and standard deviation were calculated to understand the numerical characteristics of the dataset.



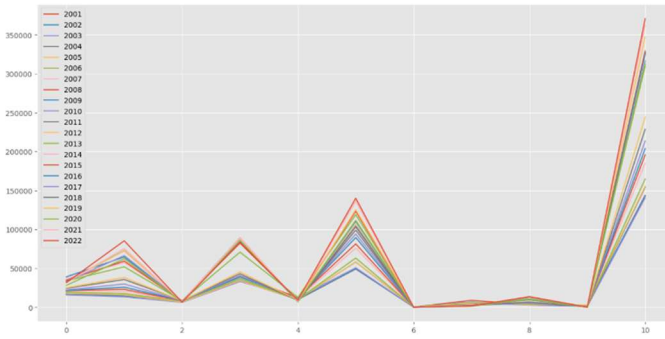
Crime Distribution Analysis

The distribution of different categories of crimes against women was analysed using bar charts and frequency plots. This analysis helped identify the most frequently reported crime categories and provided insights into the overall crime pattern within the dataset.



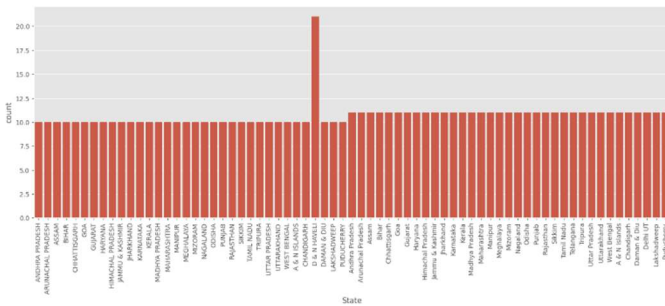
Year-wise Trend Analysis

A temporal analysis was performed to examine changes in reported crime cases over the years. Line charts were used to visualize increasing or decreasing trends, enabling a better understanding of how crime patterns evolved over time.



State-wise Crime Analysis

The geographical distribution of crime cases was analysed by comparing reported cases across different states and union territories. This analysis helped identify regions with relatively higher crime rates and highlighted geographical variations in crime occurrence.

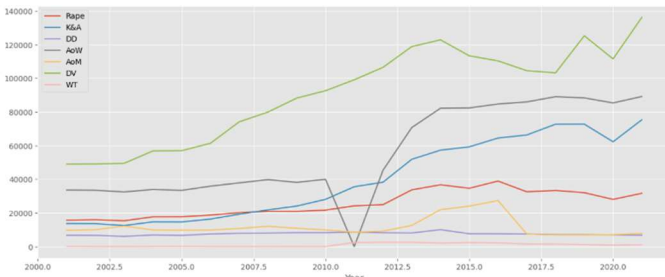


Correlation Analysis

Correlation analysis was conducted to measure the relationships among numerical variables in the dataset. A correlation matrix and heatmap were generated to identify positively or negatively correlated features, assisting in feature selection and improving model performance.

Data Visualization

Various visualization techniques, including histograms, bar charts, line plots, box plots, and heatmaps, were used to represent the dataset graphically. These visualizations simplified the interpretation of complex data and facilitated the identification of trends, distributions, and potential outliers.



Insights from EDA

The exploratory analysis revealed significant variations in crime patterns across different years and regions. It also identified important features that could influence predictive modelling. These findings provided valuable guidance for

feature selection and the development of accurate machine learning models.

V. MACHINE LEARNING MODELS

To predict domestic violence-related crime patterns, multiple supervised machine learning algorithms were implemented and compared. The objective was to evaluate the performance of different classification techniques and identify the most suitable model for accurate prediction. The models were trained using the preprocessed dataset and evaluated using standard performance metrics, including Accuracy, Precision, Recall, and F1-Score.

A. Logistic Regression

Logistic Regression is a widely used supervised machine learning algorithm for classification problems. It estimates the probability of a given input belonging to a particular class by applying the logistic (sigmoid) function. In this research, Logistic Regression was implemented as a baseline model to establish a reference for comparing the performance of other machine learning algorithms. Although computationally efficient and easy to interpret, its performance may decrease when the relationship between features and the target variable is highly non-linear.

B. Decision Tree

Decision Tree is a supervised learning algorithm that classifies data by recursively splitting the dataset based on the most informative features. The model constructs a tree-like structure consisting of decision nodes and leaf nodes, making it highly interpretable. In this study, the Decision Tree algorithm was applied to identify significant patterns within the crime dataset. However, Decision Trees are susceptible to overfitting when trained on complex datasets without appropriate pruning techniques.

C. Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting. Each tree is trained using a random subset of the training data, and the final prediction is determined through majority voting. Due to its robustness and ability to handle complex datasets, Random Forest demonstrated strong predictive performance in this research and served as one of the primary models for comparison.

D. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised classification algorithm that identifies an optimal decision boundary, known as the hyperplane, to separate different classes. It is particularly effective for high-dimensional datasets and provides strong generalization capability. In this study, SVM was implemented to evaluate its effectiveness in predicting domestic violence-related crime patterns and to compare its performance with other classification models.

E. Extreme Gradient Boosting (XGBoost)

XGBoost is an advanced ensemble learning algorithm based on gradient boosting. It constructs multiple decision trees

sequentially, where each new tree minimizes the errors of the previous trees. XGBoost is known for its high accuracy, computational efficiency, and ability to handle complex relationships within data. In this research, XGBoost was implemented as one of the advanced predictive models and compared with the other machine learning algorithms to identify the most effective approach.

F. Comparative Model Evaluation

All implemented models were trained using the preprocessed dataset and evaluated on the testing dataset. Their performance was assessed using Accuracy, Precision, Recall, and F1-Score. The comparative evaluation enabled the identification of the best-performing model based on predictive accuracy and overall classification performance. The selected model provides a reliable approach for analyzing domestic violence-related crime data and supporting data-driven decision-making.

TABLE 4 (Models ML)

Model	Type	Purpose	Advantages
Logistic Regression	Classification	Baseline prediction model	Simple, interpretable, computationally efficient
Decision Tree	Classification	Pattern identification	Easy to interpret and visualize
Random Forest	Ensemble Classification	Improved prediction accuracy	Reduces overfitting and handles complex data
Support Vector Machine (SVM)	Classification	Class separation	Effective for high-dimensional data
XGBoost	Ensemble Boosting	High-performance prediction	High accuracy and efficient learning

VI. RESULT AND PERFORMANCE EVALUATION

This section presents the experimental results obtained from the implemented machine learning models. The models were evaluated using the testing dataset to compare their predictive performance. Standard evaluation metrics, including Accuracy, Precision, Recall, and F1-Score, were used to assess the effectiveness of each model. A comparative analysis was then conducted to identify the most suitable algorithm for domestic violence prediction.

A. Evaluation Metrics

To evaluate the performance of the machine learning models, four standard classification metrics were considered.

1) Accuracy

Accuracy measures the proportion of correctly classified instances to the total number of instances in the dataset. It provides an overall measure of model performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN represent True Positive, True Negative, False Positive, and False Negative, respectively.

2) Precision

Precision indicates the proportion of correctly predicted positive observations among all predicted positive observations. A higher precision value indicates fewer false positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

3) Recall

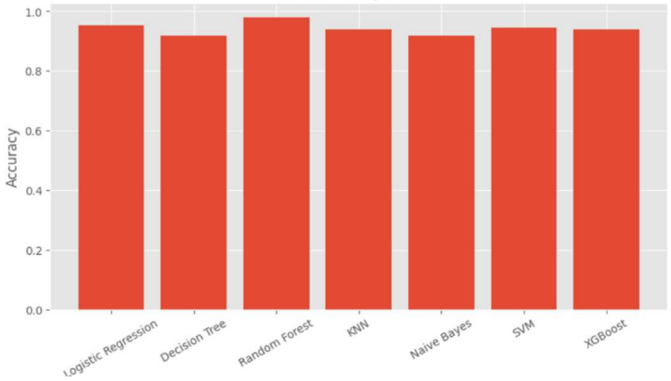
Recall measures the ability of the model to correctly identify actual positive cases. It is particularly important when minimizing false negatives is essential.

$$Recall = \frac{TP}{TP + FN}$$

4) F1-Score

The F1-Score is the harmonic mean of Precision and Recall. It provides a balanced evaluation when both false positives and false negatives are considered important.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$



B. Model Performance Comparison

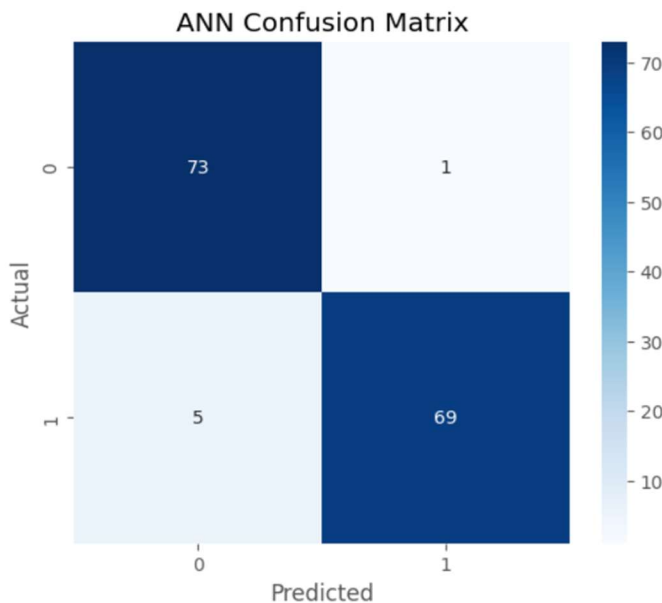
TABLE 5 (Comparative Studies)

Model	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	95.27	0.9855	0.9189	0.9510
Decision Tree	91.89	0.9559	0.8784	0.9155
Random Forest	97.97	1.0000	0.9595	0.9793
SVM	94.59	0.9853	0.9054	0.9437
XGBoost	93.92	0.9851	0.8919	0.9362

C. Confusion matrix and comparative analysis

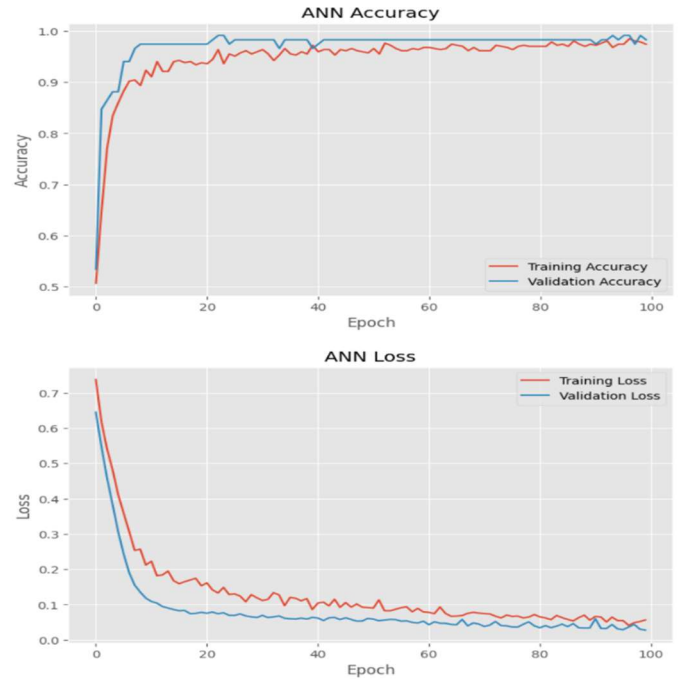
The confusion matrix was used to visualize the classification performance of each model. It illustrates the number of correctly and incorrectly classified instances through True Positives, True Negatives, False Positives, and False Negatives. A higher number of correct classifications indicates better predictive performance.

The experimental results indicate that all implemented machine learning algorithms successfully learned patterns from the dataset. However, differences in predictive performance were observed across the models. Ensemble-based algorithms demonstrated more stable and reliable performance compared to individual classifiers because of their ability to reduce overfitting and capture complex relationships within the data. Comparative evaluation enabled the identification of the most suitable model for domestic violence prediction based on the selected evaluation metrics.



D. Best Model Selection

Random Forest is selected as the best-performing model.



VII. SYSTEM IMPLEMENTATION

The proposed framework was implemented using Python in a Jupyter Notebook environment. The implementation involved data preprocessing, exploratory data analysis, feature engineering, machine learning model development, and performance evaluation. The dataset was first cleaned and transformed into a suitable format before training multiple machine learning algorithms. The trained models were then evaluated using standard classification metrics, and the best-performing model was selected for prediction.

A. Software Tools

- Python 3.x
- Jupyter Notebook
- Anaconda Distribution

B. Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- TensorFlow/Keras
- Joblib

C. Implementation Steps

- Collection of domestic violence and crime-related datasets.
- Data cleaning and preprocessing, including handling missing values and encoding categorical variables.
- Exploratory Data Analysis (EDA) to understand trends and relationships within the dataset.
- Feature engineering and train-test data splitting.
- Training of multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost.
- Evaluation of each model using Accuracy, Precision, Recall, and F1-Score.
- Selection and saving of the best-performing model using Joblib for future prediction.
- Incorporate additional socio-economic, demographic, and behavioural factors that may influence domestic violence patterns.
- Develop an explainable AI (XAI) framework to improve the transparency and interpretability of model predictions.
- Extend the framework to support real-time data analysis and continuous model updates using newly available crime records.
- Develop a web-based or mobile application to enable easy access for researchers, policymakers, and law enforcement agencies.
- Validate the proposed framework using datasets from different regions to evaluate its scalability and robustness across diverse populations.

VIII. LIMITATIONS

Although the proposed framework achieved high prediction accuracy, several limitations remain.

- The performance of the model depends heavily on the quality and completeness of the available dataset.
- The dataset represents historical crime records and may not capture recent trends or newly emerging patterns.
- The prediction results are based only on the selected features available in the dataset; important social, economic, or psychological factors are not included.
- The current implementation focuses on offline analysis and prediction and does not support real-time data processing.
- The proposed framework was evaluated using a limited set of machine learning algorithms, and other advanced deep learning approaches may further improve performance.
- The model has been validated only on the available dataset; its performance may vary when applied to different regions or unseen datasets.

IX. FUTURE WORK

Although the proposed framework achieved high predictive performance, several improvements can be incorporated in future research to enhance its applicability and effectiveness.

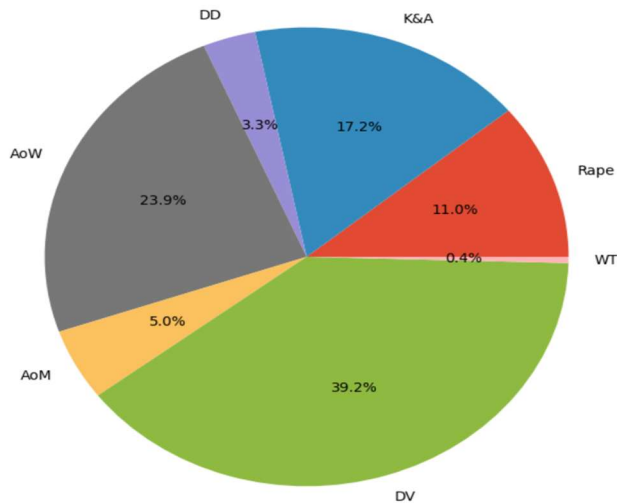
- Integrate larger and more diverse datasets from multiple sources to improve the model's generalization capability.
- Explore advanced deep learning architectures, such as Long Short-Term Memory (LSTM), Graph Neural Networks (GNN), or Transformer-based models, for improved prediction performance.

X. RESULT AND CONCLUSION

This research presented a machine learning-based framework for the analysis and prediction of domestic violence using crime-related datasets. The proposed methodology incorporated data preprocessing, exploratory data analysis, feature engineering, and comparative evaluation of multiple supervised machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost.

Experimental results demonstrated that all implemented models achieved satisfactory predictive performance. Among them, the **Random Forest** classifier produced the best results, achieving an **Accuracy of 97.97%**, **Precision of 1.0000**, **Recall of 0.9595**, and an **F1-Score of 0.9793**, making it the most effective model for the given dataset. The comparative analysis confirmed the advantages of ensemble learning techniques in handling complex crime-related data and improving prediction accuracy.

Overall, the proposed framework provides an effective and data-driven approach for analysing domestic violence patterns and supporting predictive decision-making. The findings of this research demonstrate the potential of machine learning techniques to assist researchers, policymakers, and law enforcement agencies in understanding crime trends and facilitating evidence-based strategies for domestic violence prevention and analysis.



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